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Enhanced heavy metal stability of sludge anaerobic digestion by different dosages of air nanobubble water

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ABSTRACT

Nanobubbled water (NBW) can improve the performance of sludge anaerobic digestion (AD) to enhance digestate dewaterability. However, the effects of different doses of NBW on the stability of heavy metals (HMs) remain unknown. This study found that NBW increased the ultimate methane production of sludge AD by 11.7 %–16.0 %. Compared with the corresponding control groups, the 40 % Air-NBW group (A40) presented the best improvement in methane production and digestate dewaterability of the three different dose groups (20 %, 40 %, and 60 % NBW: mixed substrates). NBW contributed to decreasing the proportion of HMs in the unstable exchangeable/acid-soluble fraction and promoting the conversion of heavy metal ions to stable low-valence states and low-energy levels. These low-valence states and low-energy levels of HMs were positively correlated with digestate dewaterability and methane production. NBW reduced the potential ecological risk index (RI) of HMs by 4.2 %–42.0 %, with the A40 showing the greatest reduction in ecological risks. The RI of HMs was negatively correlated with methane production and digestate dewaterability. NBW can simultaneously improve methane production, digestate dewaterability, and heavy metal stability in sludge AD, with 40 % NBW being the optimal dose. This study provides a new sludge treatment strategy for controlling HMs pollution.

1. Introduction

The global production of sewage sludge is projected to reach approximately 130 million tons by 2030 (with a moisture content of 80 %) [1]. As the final product of municipal sewage treatment plants, dewatered sludge (DS) is rich in biodegradable organic matter, including proteins, polysaccharides, and lignocellulose, which are advantageous for agricultural applications after treatment [2]. However, DS also contains certain heavy metals (HMs), which can significantly diminish its usability and pose serious risks to environmental and ecological security [3]. Anaerobic digestion (AD) has emerged as the predominant technology for treating DS [4]. This process not only generates clean energy and reduces sludge volume but also stabilizes HMs within the sludge to some extent [2,5]. Nevertheless, conventional AD still encounters several challenges, such as low methane production, poor dewaterability, and limited stabilization of HMs [6].

To improve sludge AD performance and enhance the stability of HMs, exogenous additives have been widely employed [6,7]. Nanoscale zero-valent iron used in the sludge mesophilic AD process increased cumulative methane production by 45.8 % and simultaneously increased the residual states of zinc (Zn) (from 15.1 % to 27.5 %) and nickel (Ni) (from 15.1 % to 26.0 %) [7]. Using biochar for sludge AD increased the oxidizable (from 23.0 % to 27.0 %) and residual (from 9.5 % to 17.5 %) states of lead (Pb) [6]. However, these approaches may inadvertently contribute to secondary environmental pollution, thereby restricting the subsequent agricultural utilization of digestate [8]. In

Abbreviations: AD, Anaerobic digestion; BCR, European Community Reference Bureau's sequential extraction procedure; Cr, Chromium; CST, Capillary suction time; Cu, Copper; DS, Dewatered sludge; E_r, Potential ecological risk factor of single metal; FAN, Free ammonia nitrogen; F1, Exchangeable/acid-soluble fraction; F2, Reducible fraction; F3, Oxidizable fraction; F4, Residual fraction; HMs, Heavy metals; HMI, Heavy metal ions; Mn, Manganese; NBW, Nanobubble water; Ni, Nickel; Pb, Lead; RI, Potential ecological risk index; SCOD, Soluble chemical oxygen demand; SRF, Specific resistance to filtration; TAN, Total ammonia nitrogen; T_r, Toxicity reaction factor; TS, Total solids; VFA, Volatile fatty acid; VS, Volatile solids; XPS, X-ray photoelectron spectroscopy; Zn, Zinc.

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Table 1
Parameters of seed sludge and dewatered sludge.

	Seed sludge	Dewatered sludge
TS (%)	10.2 ± 0.2	15.8 ± 0.2
VS (%)	2.1 ± 0.1	9.5 ± 0.2
VS/TS (%)	20.9 ± 0.3	60.3 ± 0.3
pH	7.24 ± 0.01	7.02 ± 0.01
SCOD (mg L^{-1})	1541.9 ± 22.6	3712.7 ± 29.8
TAN (mg L^{-1})	1418.2 ± 70.5	651.0 ± 24.5
FAN (mg L^{-1})	31.1 ± 0.8	8.6 ± 0.4
C (%, dry solids)	9.1 ± 0.1	30.5 ± 0.3
N (%, dry solids)	1.2 ± 0.0	5.3 ± 0.1
H (%, dry solids)	1.3 ± 0.0	4.3 ± 0.1
S (%, dry solids)	0.6 ± 0.0	0.6 ± 0.0
C/N	7.6 ± 0.1	5.8 ± 0.1

Note: All values are expressed as mean ± standard deviation ($n = 3$).

addition, the improved dewaterability can reduce the toxicity and potential risks of HMs in DS [9,10]. Consequently, an environmentally friendly and economically viable comprehensive strategy needs to be proposed to enhance the stabilization of HMs based on improved methane production performance and digestate dewaterability.

Nanobubble water (NBW) contains a substantial number of bubbles with diameters <1 μm [11]. Due to the high pressure, small diameter, and large specific surface area of nanobubbles, NBW exhibits properties not found in conventional water. These properties include high gas solubility, strong adsorption, extended storage time, and high zeta potential [12]. The hydroxyl radicals generated upon the collapse of nanobubbles in Air-NBW are highly oxidizing and contribute to the decomposition of complex substrates, leading to a 19.1 % increase in methane production in the AD system [13]. Air-NBW can effectively alleviate ammonia inhibition by stimulating the activities of coenzyme F_{420} and hydrolase, resulting in a 12 %–39 % increase in methane production [14]. Furthermore, the high surface activity of nanobubbles can alter the charge distribution on the surface of sludge particles, thereby enabling the release of bound water from improved flocs. Different types of NBW (Air, CO_2, He, and N_2) have been shown to reduce the specific resistance to filtration (SRF) and capillary suction time (CST) of digestate by 18.8 %–29.6 % and 14.6 %–18.2 %, respectively [15]. However, the effects of different doses of NBW on methane production, digestate dewaterability, and the stability of HMs in sludge AD have been rarely investigated. In addition, the correlation between methane production, digestate dewaterability, and the stability of HMs remains unknown.

This study investigated the effects of different dosages of Air-NBW on methane production, microbial communities, and dewaterability of DS. The speciation and valence changes of HMs by Air-NBW were further explored, and the potential ecological risk index (RI) and the potential ecological risk factor of single metal (E_r) were used to comprehensively assess the potential of Air-NBW to mitigate the ecological risks of HMs. Finally, the stability and ecological risks of HMs in sludge AD were analyzed in relation to methane production and digestate dewaterability. This study contributes to developing an efficient strategy for resource recovery while controlling HMs pollution.

Table 2
Total solids, volatile solids, and substrate-to-inoculum ratio of different groups.

Group	TS (%)	VS (%)	S/I ratio (on the VS basis)
A/C20	12.4 ± 0.2	7.1 ± 0.2	36.2 ± 1.6
A/C40	9.5 ± 0.2	5.4 ± 0.1	27.2 ± 1.2
A/C60	6.7 ± 0.2	3.6 ± 0.1	18.1 ± 0.8

Note: All values are expressed as mean ± standard deviation (n = 3).

2. Materials and methods

2.1. Materials

DS served as the substrate, which was obtained from a sewage treatment plant, employing an "oxidation ditch + biological aerated filter" process with a 6000 m^3 d^{-1} treatment capacity. The seed sludge was taken from a mesophilic anaerobic digester in the laboratory. Table 1 shows the main parameters of different sludge.

2.2. Experimental process

The preparation of Air-NBW was consistent with Wang et al. [15], and the average particle size and concentration of nanobubbles were 135.8 nm and 537 million mL^{-1} (Fig. S1). The anaerobic digesters were consistent with Xu et al. [4] (Fig. 1). DS was mixed with Air-NBW or deionized water in different mass ratios (water: DS = 2: 8, 4: 6, and 6: 4), and then they were mixed with seed sludge (10: 1), with a total amount of 1500 g. According to the mass ratios of Air-NBW or deionized water to the mixture of water and DS, which were 20 %, 40 %, and 60 %, the study was divided into six groups, labeled A20, C20, A40, C40, A60, and C60. Table 2 shows the total solids (TS), volatile solids (VS) and substrate-to-inoculum (S/I) ratio of the groups. Each group performed three parallel experiments with a total of 18 digesters. The water bath

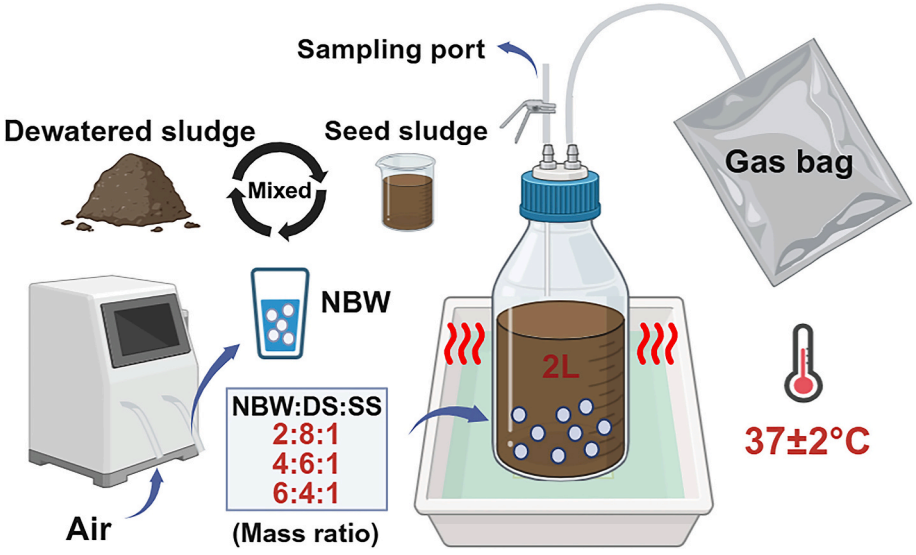


Fig. 1. Experimental setup for anaerobic digestion.

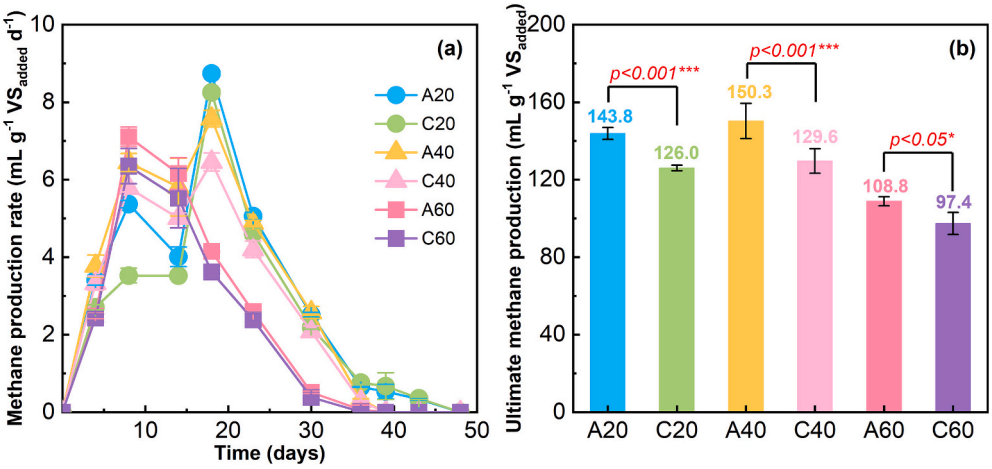


Fig. 2. Changes in (a) methane production rate and (b) ultimate methane production during anaerobic digestion.

temperature was 37 ± 2 °C, and the digestion period was 43 days.

2.3. Analytical methods

TS, VS, and pH of solid samples, soluble chemical oxygen demand (SCOD), and ammonia of liquid samples were measured using standard methods [16], and free ammonia nitrogen (FAN) of total ammonia nitrogen (TAN) was calculated using the specific calculation formula [17]. The gas chromatograph was used to determine the volatile fatty acid (VFA) of liquid samples (SP-7890Plus, Rui Hong, China), the acid gas flow meter was used to determine the biogas volume (LML-2, Alpha, China), and the biogas composition analyzer was used to determine the methane concentration (Biogas 5000, Geotech, UK). Methane production was fitted using the modified Gompertz formula [13]. SRF measurement methods were consistent with Yang et al. [18]. CST was measured with a capillary absorption meter (304 M, Triton Electronics, UK). The laser particle size analyzer was used to determine the digestate particle size (Nano ZS90, Malvern, UK).

The inductively coupled plasma emission mass spectrometer was used to quantify the concentration of HMs (ICP-MS 7850, Agilent, USA). The European Community Reference Bureau's (BCR) sequential extraction procedure was used to determine the exchangeable/acid-soluble fraction (F1), the reducible fraction (F2), the oxidizable fraction (F3), and the residual fraction (F4) of HMs. The F1 + F2 fractions are the bioavailability of HMs, expressed as the degree to which metal elements can interact with biological systems, and can be used to evaluate the toxicity of HMs [19]. The specific operation methods were consistent with Guo et al. [20]. X-ray photoelectron spectroscopy (XPS) was used to analyze the valence state of HMs (Escalab 250Xi, Thermo, USA). Finally, the E_r and RI were calculated by the formulas to evaluate the environmental risks of HMs [21], in which the toxicity reaction factors (T_r) of copper (Cu), Ni, Zn, Pb, manganese (Mn), and chromium (Cr) were 5, 5, 1, 5, 1, and 2, respectively.

DNA was extracted from samples on day 8, and the high-throughput sequencing technology was used to analyze the microbial communities after adding different NBW. The V3-V4 portions of the bacterial 16S rRNA gene were amplified using primers 341F (5 '-CCTACGGGAGGCAGCAG-3') and 806R (5 '-GGACTACHVGGGTWTCTAAT-3'); The V4-V5 portions of the archaeal 16S rRNA gene were amplified using primers Arch519F (5 '-CAGCCGCCGCGGTAA-3') and Arch915R (5 '-GTGCTCCCCCGCCAATTCCT-3'). The sequencing data has been submitted online (https://www.ncbi.nlm.nih.gov). The accession numbers were PRJNA1236884 and PRJNA1236892. Prediction of genes from the sequencing data using PICRUSt2 to analyze the potential of key enzymes in microbial communities (https://github.com/picrust/picrust2). The kyoto encyclopedia of genes and genomes databases were used to analyze functional genes (https://www.kegg.jp/).

2.4. Statistical analysis

The statistical analysis was performed by SPSS (27.0) software. One-way analysis of variance (ANOVA) was performed to test the differences between different experimental groups, and $p < 0.05$ means statistical significance. The correlation between the two parameters was analyzed by Spearman's correlation coefficient, and * indicates $p < 0.05$, ** indicates $p < 0.01$, and *** indicates $p < 0.001$. All data are expressed as mean ± standard deviation.

3. Results and discussion

3.1. Digestion performance and microbial communities

On day 4, the VFA levels in the NBW groups increased by 10.5 %–20.3 % compared with the control (Fig. S2). NBW enhanced microbial activity and improved the hydrolysis performance of sludge AD [12]. On day 43, A20 and C20, which had higher solid content, exhibited VFA accumulation, potentially leading to acid inhibition and reduced methane production [22]. In this study, all groups showed high S/I ratios (>10), with A20 and C20 exhibiting the highest values (Table 2). It was found that the increased S/I ratio would accelerate the rate of hydrolysis and acidification, enhance the formation of organic acid intermediates during AD, increase the risk of acid accumulation, and adversely affect methane production [23]. The pH, SCOD, TAN, and FAN are illustrated in Fig. S3. Compared with the control, the TAN in the NBW groups increased by 13.3 %–34.7 % ($p < 0.01$) on day 4. The collapse of nanobubbles releases chemical energy while generating hydroxyl radicals, which have strong oxidizing properties and contribute to the hydrolysis of proteins [24]. The rigid cell walls of microorganisms in sludge often reduce the efficiency of anaerobic degradation and prolong the hydraulic retention time, thereby affecting hydrolysis efficiency [25]. Nanobubbles can create localized high pressure and microjets upon collapse that may destroy the cell walls of microorganisms (e.g., peptidoglycan layers) [26]. In addition, the hydroxyl radicals generated by nanobubbles may oxidize lipids, proteins, and polysaccharides in the cell wall, destroying its integrity [12,27]. Once the cell wall is damaged, intracellular substances (e.g., proteins and polysaccharides) are released and further degraded, promoting hydrolysis and leading to increases in TAN and VFA [25]. This was confirmed by the elevation of VFA and TAN by the addition of NBW on day 4. In the later stage of AD, with the metabolism of VFA and the increase of TAN, the pH gradually showed an upward trend. The alkaline environment favors the growth of hydrogenotrophic methanogens [28]. The FAN of

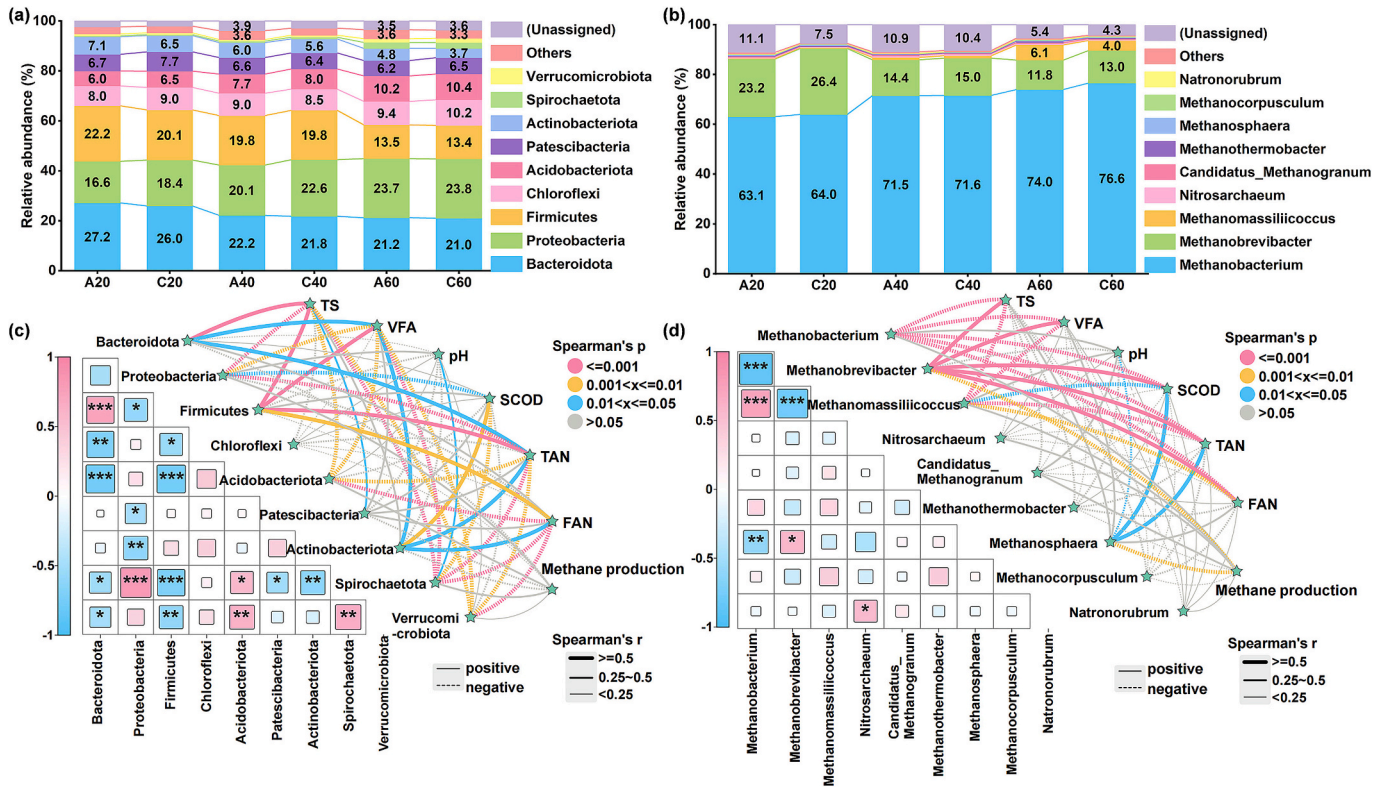


Fig. 3. The relative abundance of (a) bacteria and (b) archaea, correlation analysis between environmental factors and (c) bacteria and (d) archaea.

the A20 and C20 exceeded 250 mg L^{-1} after day 23, so the methanogens may be subject to significant ammonia inhibition [29].

After day 8, all groups exhibited a general downward trend, and most of the organic matter in the A60 and C60 had been converted to methane (Fig. 2a). The decline observed in the other groups should be contributed by the ammonia inhibition of methanogen activity due to TAN greater than 1500 mg L^{-1} (Fig. S3) [4]. As the activity of methanogens gradually recovered, the methane production rate peaked on day 18. Among the control groups, the ultimate methane production of the C40 surpassed that of the others (Fig. 2b and Table S1). For the C20, high TS may have been influenced by FAN toxicity (Fig. S3), which could compromise the stability of the anaerobic system [14]. Furthermore, VFA accumulation might also lead to acid inhibition (Fig. S2) [22]. For the C60, when the TS was low, to meet the growth needs of microorganisms, some organic matter in the substrate was forced to synthesize new hydrolytic bacteria and methanogens, thus limiting methane production [30]. It was found that with the increase of sludge TS, methane production first increased and then decreased [30,31]. During sludge AD with different TS, the cumulative methane production in descending order was TS = 8 %, TS = 6 %, and TS = 12 % [32], which was similar to the trend observed in this study (Table 2). NBW significantly enhanced the methane production by 14.1 % ($p < 0.001$), 16.0 % ($p < 0.001$), and 11.7 % ($p < 0.05$) for the A20, A40, and A60, respectively, compared with their corresponding controls. NBW promotes enzyme activity and increases electron transfer to stimulate the activity of methanogens, thus improving methane production [12]. The lower content of nanobubbles in the A20 resulted in a weaker promoting effect compared with A40. Furthermore, since the control group had the optimal hydrolysis conditions, it limited the improvement efficiency of NBW on hydrolysis performance in the A60, consequently making the promoting effect of methane production less than that of the A40 [15].

At the phylum level, *Bacteroidota* (21.0 %–27.2 %), *Proteobacteria* (16.6 %–23.8 %), *Firmicutes* (13.4 %–22.2 %), *Chloroflexi* (8.0 %–10.2 %), and *Acidobacteriota* (6.0 %–10.4 %) dominated the bacterial community (Fig. 3a). *Bacteroidota* and *Firmicutes* are known to participate in the hydrolysis of lignocellulose to generate short-chain fatty acids [4]. Compared with the control, the relative abundance of *Bacteroidota* and *Firmicutes* increased in the NBW groups. In addition, high S/I conditions can promote the activity of hydrolytic and acidogenic bacteria, thereby enhancing the hydrolysis performance [23]. In this study, the relative abundance of *Bacteroidota* and *Firmicutes* increased with increasing S/I ratios. At the genus level, *Methanobacterium* (64.0 %–76.6 %), *Methanobrevibacter* (11.8 %–26.4 %), and *Methanomassiliicoccus* (0.6 %–6.1 %) dominated the archaeal community (Fig. 3b). These genera are all hydrogenotrophic methanogens [15]. Hydrogenotrophic methanogens demonstrate strong tolerance under high ammonia, which has certain advantages in maintaining the methane production rate [33]. High S/I conditions can increase the activity of hydrogenotrophic methanogens, thereby promoting methane production [34]. Therefore, the hydrogenotrophic methanogenesis pathway was dominant in this study. The relative abundances of *Methanobacterium* and *Methanobrevibacter* in the NBW groups were reduced by 0.1 %–2.6 % and 0.6 %–3.2 %, respectively, compared with the control. This reduction suggests a shift in the archaeal community structure, indicating that more types of methanogens were involved in the methanogenesis process. This implies that NBW stimulated the activity of more types of archaea, which is beneficial to the increase in the diversity of the archaea community and makes methanogenesis metabolism active [35]. The abundance changes of key enzymes in major methanogenesis pathways are presented in Fig. S4. The abundance of key enzymes involved in the M00567 pathway was increased in the NBW groups, compared with the control. NBW made hydrogenotrophic methanogenesis metabolism more active.

3.2. Dewaterability

Particle size (d50) is an important factor affecting digestate dewaterability [36]. Smaller particles possess a larger specific surface area,

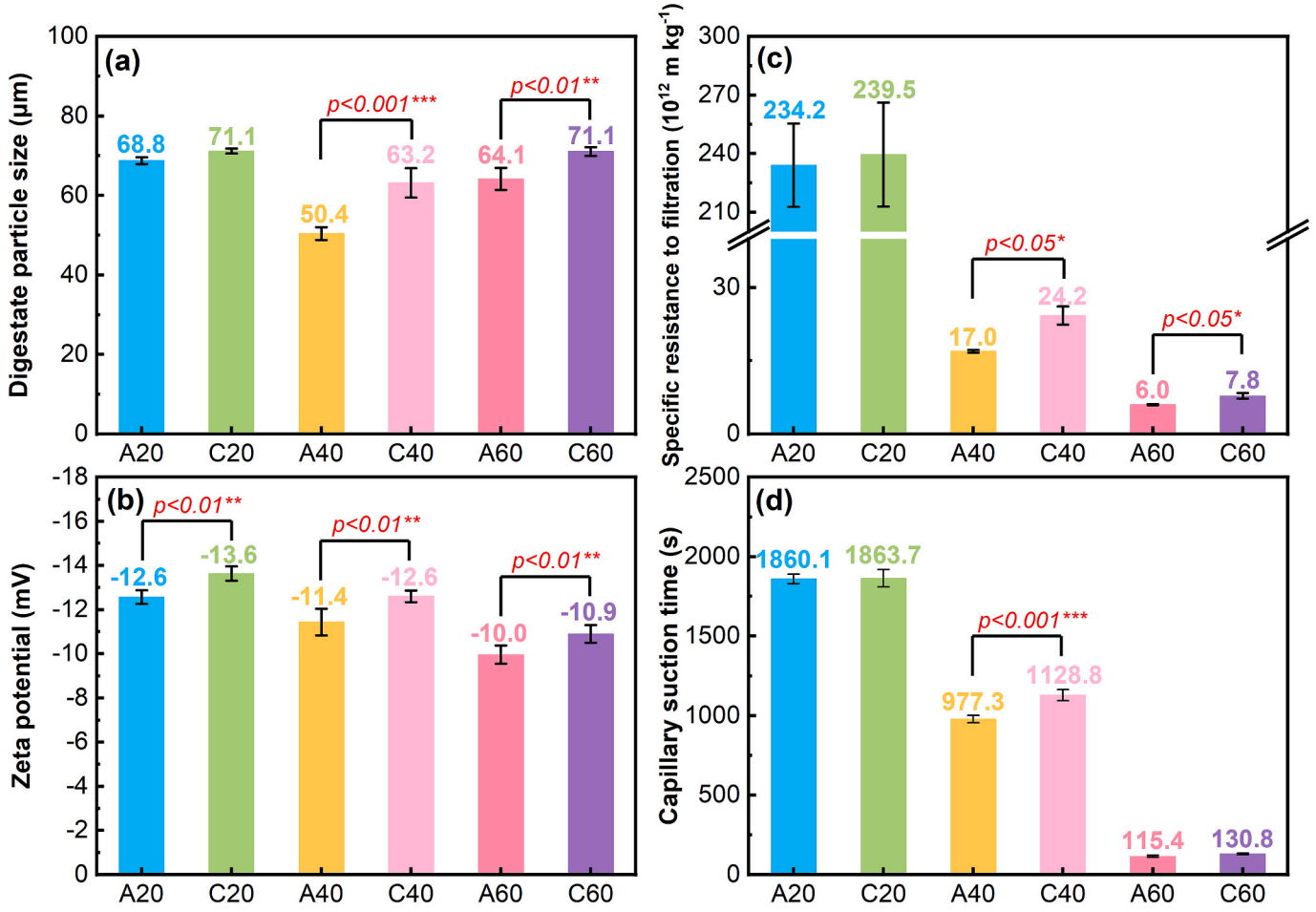


Fig. 4. Changes in (a) digestate particle size, (b) zeta potential, (c) specific resistance to filtration, and (d) capillary suction time.

facilitating the separation of water [37]. Lower TS levels enhance mass transfer efficiency and promote a more complete degradation of organic matter within the sludge [30]. However, the d50 of C60 exceeded that of C40 (Fig. 4a). This may be because small particles of sludge are more readily utilized, while large particles of sludge tend to remain under low organic loading conditions, thereby increasing the average particle size of digestate after AD [4]. The d50 of the A20, A40, and A60 decreased by 3.4 %, 20.1 % ($p < 0.001$), and 9.7 % ($p < 0.01$), respectively, compared with their corresponding controls. Nanobubbles enhanced the mass transfer efficiency, improved the hydrolysis performance, and decreased the particle size of digestate, thereby enhancing the dewaterability [15,38]. Among the NBW groups, A40 exhibited the best improvement in hydrolysis performance. The TAN improved by 4.1 %, 7.5 %, and 5.2 % for A20, A40, and A60, respectively, compared with the control on day 43 (Fig. S3). Consequently, A40 demonstrated the most pronounced reduction in d50.

Zeta potential is the crucial index to evaluate the surface electronegativity of digestate [36]. With the increase in zeta potential, the collision of digestate particles is promoted, leading to the release of bound water from the improved digestate flocs and enhancing digestate dewaterability [15]. Lower TS improves the degradation of digestate flocs and reduces electrostatic repulsion between particles, resulting in increased zeta potential (Fig. 4b) [39]. The Zeta potential of the A20, A40, and A60 increased by 6.9 % ($p < 0.01$), 9.2 % ($p < 0.01$), and 8.5 % ($p < 0.01$), respectively, compared with the control. Nanobubbles can attract and immobilize cations in the digestate through electrostatic interaction to neutralize the negative charges on the surface of digestate particles, thereby reducing electrostatic repulsion on the surface and increasing the zeta potential [40]. Although lower TS generally leads to more complete organic matter degradation, A40 exhibited the most significant increase in zeta potential among the NBW groups. This may be because a higher dosage of NBW can lead to excessive aggregation of nanobubbles within the digestate, potentially disrupting the floc structure and consequently increasing the electrostatic repulsion of the flocs to some extent [12,39]. Therefore, an appropriate dosage of NBW is more conducive to enhancing dewaterability.

SRF can reflect the consistency and structural characteristics of sludge, and lower SRF values are favorable for microbial infiltration and degradation [41]. CST refers to the time required to measure the permeation distance of moisture in sludge on filter paper under specific conditions [41]. Digestate particle size and Zeta potential can indirectly reflect the changes in SRF and CST by affecting flocs' structure and porosity. Organic matter degradation in AD led to a decrease in the particle size of digestate and reduced the electrostatic repulsion between particles, which facilitates the aggregation of digestate particles into larger flocs, increasing the porosity of digestate, thus leading to the enhancement of digestate permeability [39,42]. The SRF of the A20, A40, and A60 decreased by 2.2 %, 29.8 % ($p < 0.05$), and 23.1 % ($p < 0.05$), respectively, compared with the control (Fig. 4c). The CST of the A20, A40, and A60 decreased by 1.9 %, 13.4 % ($p < 0.001$), and 11.8 %, respectively, compared with the control (Fig. 4d). The most significant reduction of SRF and CST was observed in A40, aligning with the results of digestate particle size and Zeta potential. Due to its high permeability, NBW can enter into the interior of digestate particles and degrade the extracellular polymers to release bound water by using active substances released when the nanobubbles collapsed; meanwhile, under the action of the nanobubbles, the charges on the surface of particles were partially neutralized and the electrostatic repulsion between particles was

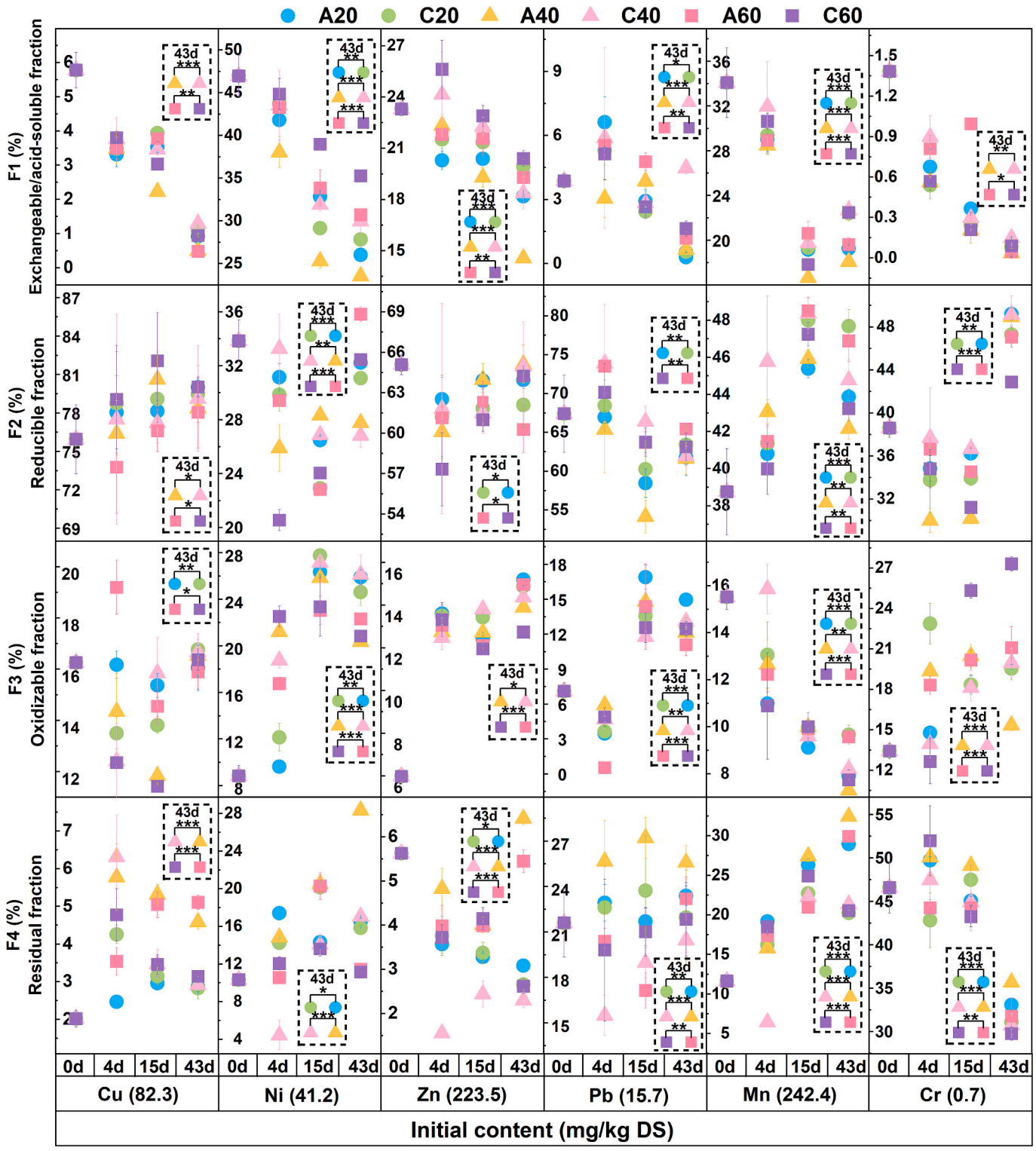


Fig. 5. Changes in speciation of different heavy metals in sludge during anaerobic digestion.

weakened, which facilitated the aggregation of the flocs [12,43]. These effects improved the floc structure, increasing porosity, which reduced SRF and CST to improve digestate dewaterability.

3.3. Speciation distribution of heavy metals

HMs are immobilized in sludge through various chemical and biological interactions, exhibiting non-degradability and certain toxicity [44]. Therefore, it is essential to mitigate the potential ecological risks by transforming HMs to a stable state [45]. According to the distribution of HMs in municipal sludge across various regions of China [46], it can be seen that compared with other HMs, the contents of cadmium, mercury, and arsenic in municipal sludge were extremely low. In the western region where this study was conducted, cadmium and arsenic were not major pollutants. In comparison, Cu, Ni, Zn, Pb, Mn, and Cr have greater research value in the municipal sludge of western China.

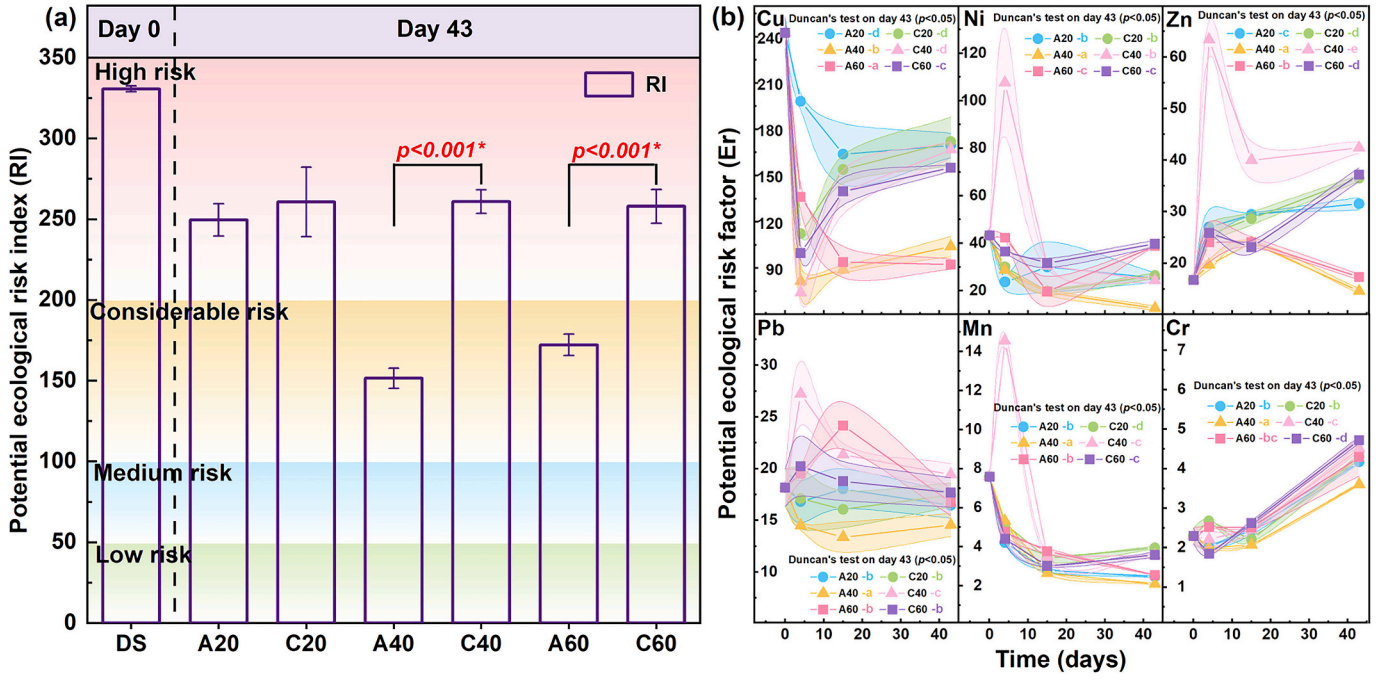


Fig. 6. (a) Potential ecological risk index in final digestate and (b) Changes in potential ecological risk factor of different heavy metals in sludge during anaerobic digestion.

The HMs speciation stability increases in the order of F1, F2, F3, and F4, with the F1 fraction being the most readily available to organisms [20].

Overall, the F1 fraction of each metal decreased after AD. The F1 fractions of Cu, Ni, Zn, Pb, Mn, and Cr were reduced to 0.9 %–1.3 %, 27.8 %–35.3 %, 18.4 %–20.4 %, 0.6 %–4.5 %, 22.4 %–22.7 %, and 0.1 %–0.1 %, respectively (Fig. 5). Generally, changes in environmental factors can directly influence the stability of HMs. The moderate increase in pH is beneficial to the stabilization of HMs [45]. During AD, the consumption of VFA and the increase of ammonia nitrogen indirectly promoted the stabilization of HMs by raising pH (Figs. S2–3). In addition, the bioavailability fractions (F1 + F2) of Cu, Ni, Zn, Pb, and Mn were reduced. The bioavailability fraction of Cr showed an increase, possibly due to the release of chlorine into metal halide formation, which converted poorly soluble metal complexes into soluble metal chlorides (e.g., $CrCl_3$) [47]. Cr is considered one of the least bioavailable metals [45], and the initial amount of Cr was minimal compared with other HMs, so the change in its bioavailability had little impact on ecological risks. The F4 fraction of Cu, Ni, Pb, and Mn all increased, while the F4 fraction of Zn slightly decreased (except for the A40). This may be because, in addition to complexing with organic functional groups, Zn also tends to react with sulfides to produce ZnS precipitates [48]. The ZnS precipitates are readily soluble under strongly oxidizing and strongly acidic conditions, so in the final step of BCR, Zn remained in the supernatant after the precipitate was dissolved, resulting in an increase in its F3 fraction [20]. In summary, AD positively impacted the stabilization of HMs (Table S2).

NBW affected the formal transformation of HMs, primarily reflected in the changes observed in each fraction of HMs. After AD, the F1 fraction of each metal in the NBW groups was reduced compared with the control. The F1 fraction of Cu, Ni, Zn, Pb, Mn, and Cr decreased by 0.2 %–0.8 %, 1.8 %–6.4 %, 1.1 %–3.8 %, 0.3 %–3.8 %, 2.9 %–4.6 %, and 0.0 %–0.1 %, respectively (subtraction of percentages). NBW possesses a high specific surface area and strong adsorption force, which may enable it to adsorb heavy metal ions (HMI) in digestate efficiently [12]. Compared with the control, the F4 fraction of each metal in the NBW groups was increased. The F4 fraction of Cu, Ni, Zn, Pb, Mn, and Cr increased by 0.0 %–2.0 %, 0.3 %–11.2 %, 0.4 %–4.1 %, 1.4 %–5.1 %, 8.7 %–11.1 %, and 2.0 %–5.0 %, respectively (subtraction of percentages). Nanobubbles are capable of enhancing the hydrolysis of complex organic matter. The products of this process (e.g., humic acid) contain abundant functional groups that can form stable complexes with HMI [45,49]. All the above results indicate that NBW is beneficial in converting HMs into a more stable form, thereby mitigating the ecotoxicity and risks of HMs, which provides convenience for subsequent digestate disposal.

Different proportions of NBW exhibited varying effects on the stabilization of HMs. Among all groups with NBW, only the bioavailability fraction (F1 + F2) of each metal in the A40 was reduced compared with the control. Compared with other proportions of groups with NBW, A40 was more conducive to stabilizing HMs. Meanwhile, the enhancement rate of the F4 fraction for each metal in the A40 surpassed that in the other groups with NBW. Notably, this is consistent with the conclusion of protein hydrolysis in AD systems. The increasing concentration of TAN was the most significant in the A40, exceeding that seen in the A20 and A60, compared with the control (Fig. S3). NBW may have stimulated the activity of microorganisms, enabling extracellular polymers produced through their metabolism to interact with HMI present in digestate, facilitating the stabilization of the HMI complex [11,50].

3.4. Risk assessment of heavy metals

E_r and RI have been widely used to evaluate the HMs' potential ecological risks in sludge substrates [21]. Generally, RI is categorized into four levels: low risk (RI < 50), medium risk (50 < RI ⟨100), considerable risk (100 < RI ⟨200), and high risk (RI > 200) [51]. Prior to AD, the RI of HMs in sludge was observed to be approximately 330, indicating a very high risk to the ecosystem (Fig. 6a). After AD, the ecological risks of each group were effectively reduced, and the RI remained around 260. Notably, the RI of the NBW groups was 4.2 %–42.0 % lower than that of the control, demonstrating that the potential environmental harm of HMs could be effectively reduced by NBW treatment. The RI of A40 and A60 fell below the threshold of 200. Among them, A40 exhibited the lowest ecological risk, which was reduced by 54.2 % compared with the RI of sludge before AD. Duncan's

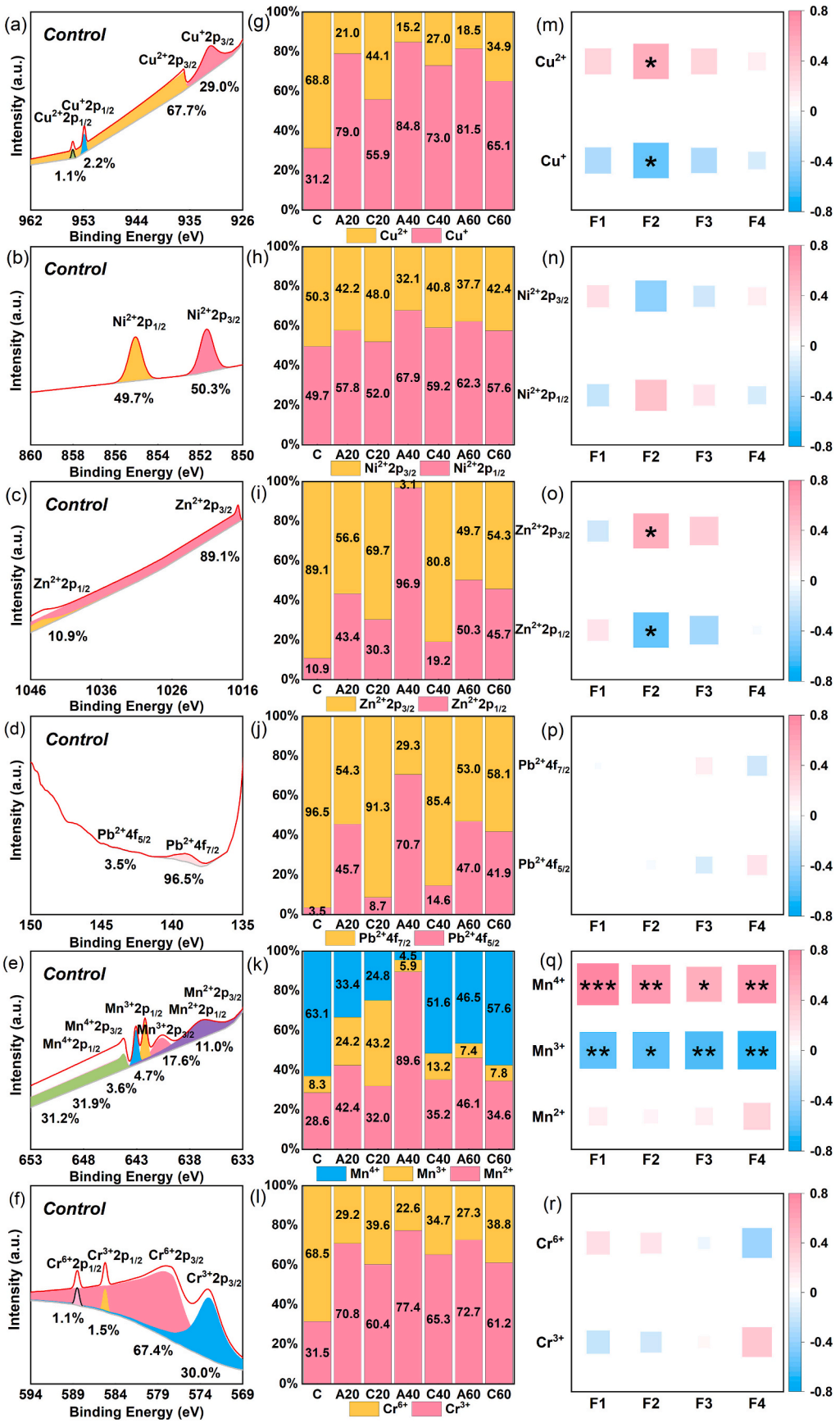


Fig. 7. X-ray photoelectron spectroscopy analysis in final digestate.

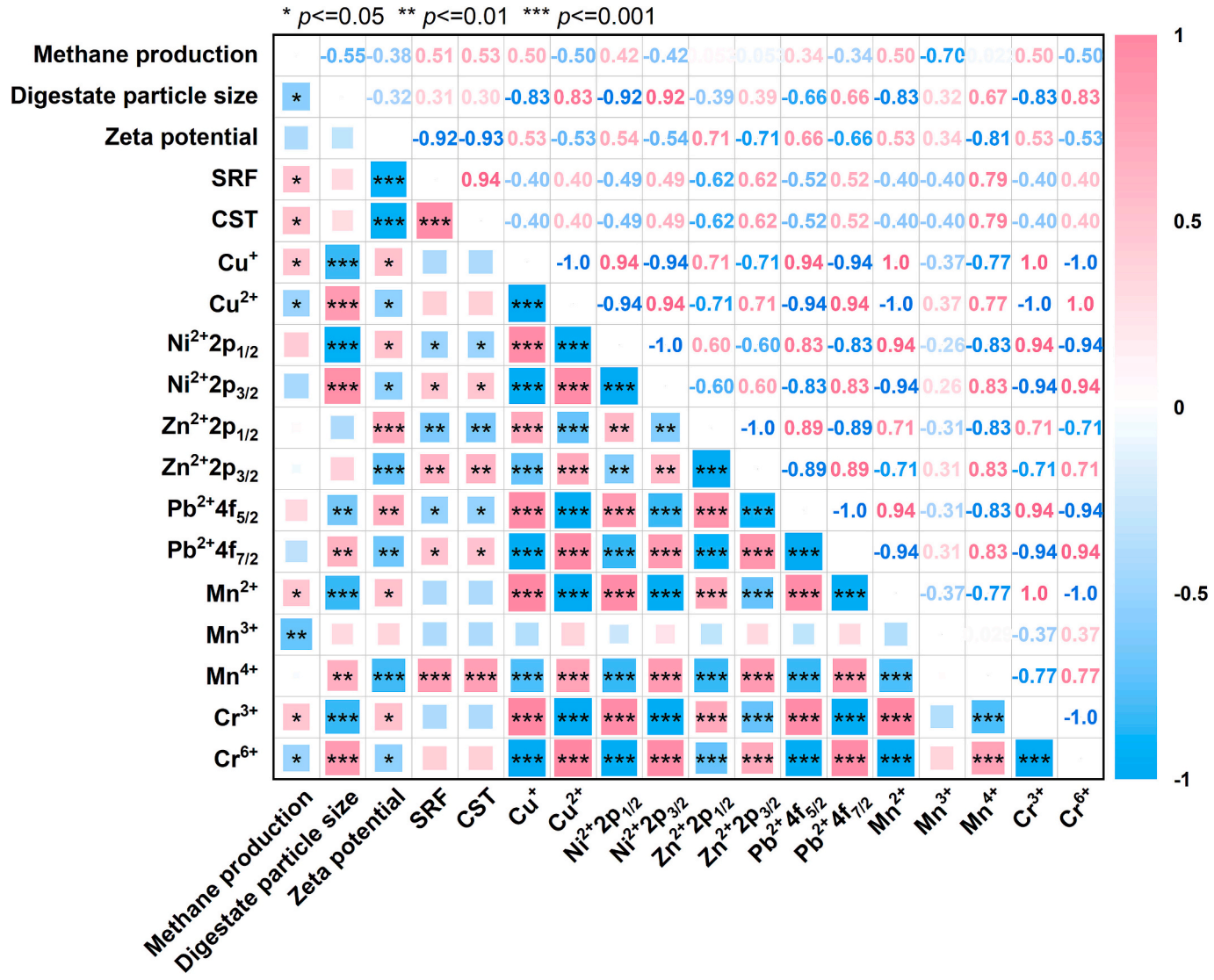


Fig. 8. Correlation analysis between heavy metals, dewaterability and methane production in final digestate.

test revealed that the Er of each metal in A40 was significantly lower than the control ($p < 0.05$) (Fig. 6b). In addition, the E_r of Cu was substantially higher than that of other HMs, suggesting that the alteration of ecological risks of HMs in digestate was mainly attributed to Cu. This can be attributed to the high content of Cu in sludge and its high toxicity response factor, resulting in a high potential ecological risk of Cu. In addition, the F4 fraction of Cu was very low (Fig. 5), making it more migratory than other HMs, which may also be one of the reasons for the high E_r of Cu.

3.5. X-ray photoelectron spectroscopy analysis

XPS can be used to analyze the chemical composition of samples and determine the valence state of elements [52]. The characteristic peaks of the binding energy of HMs can be used to identify the element species, and the shifts of binding energy reflect the change in the valence state of HMs [53].

According to the XPS results, the proportion of more toxic high-valence HMI was greater than that of low-valence HMI before AD (Figs. S5–10). The proportion of low-valence HMI gradually increased with the progression of AD. The proportion of Cu^{+} increased from 31.2 % to 55.9 %–73.0 %, and Cu^{2+} gradually converted to Cu^{+} (Fig. 7). Cu^{2+} usually exists in combination with organic matter, while Cu^{+} usually exists in combination with sulfur [54]. After AD, most of the organic matter was consumed, gradually decreasing the proportion of Cu^{2+}. Mn and Cr also exhibited the same pattern. The proportion of Mn^{2+} increased from 28.6 % to 32.0 %–35.2 %, and Mn^{4+} eventually changed to Mn^{2+} through Mn^{3+} transition. The proportion of Cr^{3+} increased from 31.5 % to 60.4 %–65.3 %, and Cr^{6+} gradually changed to Cr^{3+}. These findings indicate that AD can promote the occurrence of the reduction reaction of HMs, which gradually transforms the high-valence HMI into the low-valence state. This was further evidenced by the significant correlation of the altered valence states of Cu and Mn with their reducible state fraction (F2). The hydrogenotrophic methanogen *Methanobacterium* may promote the reduction of HMs, making them more stable [55,56]. *Methanobacterium* dominated the methanogenesis process (Fig. 3) and benefited the reduction and stabilization of HMs. Moreover, with the progress of AD, HMs gradually transformed from high energy levels to stable low energy levels. Ni^{2+} and Zn^{2+} gradually converted from $2p_{3/2}$ to $2p_{1/2}$, and Pb^{2+} gradually converted from $4f_{7/2}$ to $4f_{5/2}$. The proportions of $Ni^{2+}2p_{1/2}$, $Zn^{2+}2p_{1/2}$, and $Pb^{2+}4f_{5/2}$ increased from initial 49.7 %, 10.9 % and 3.5 % to 52.0 %–59.2 %, 19.2 %–45.7 %, and 8.7 %–41.9 %, respectively. Through the conversion of the valence state and energy level of HMI, it can be seen that AD can promote HMs to exist in digestate in a more stable and less toxic form.

After AD, the proportion of low-valent HMI in the NBW groups

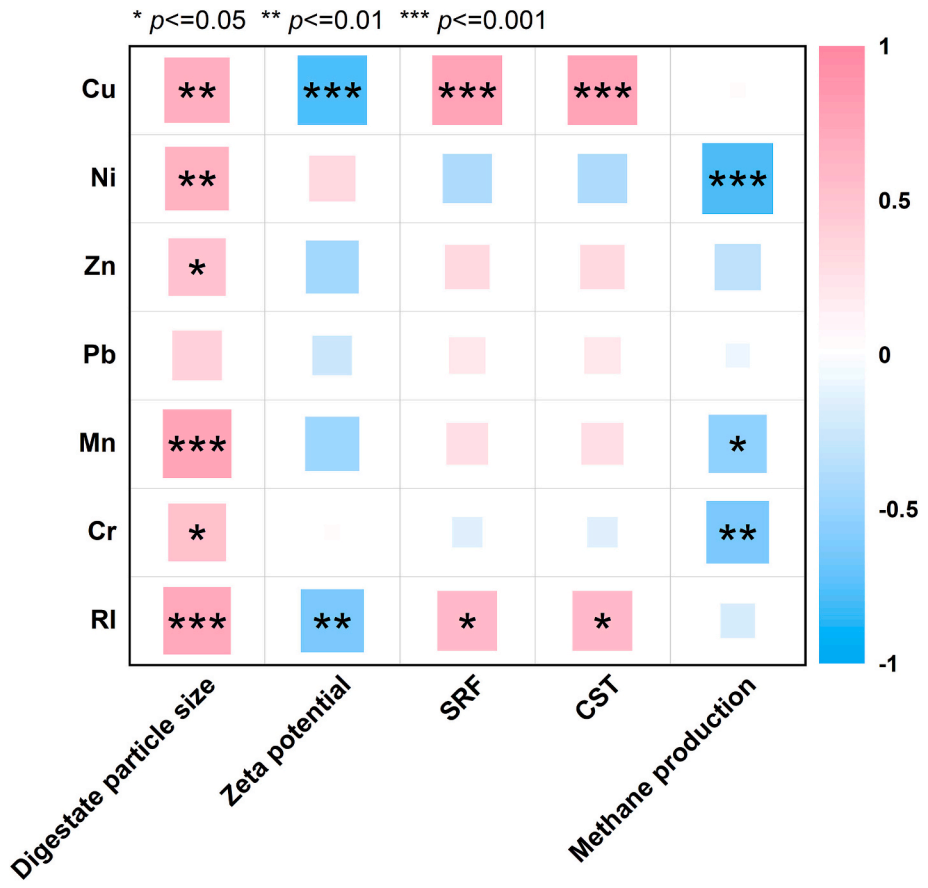


Fig. 9. Correlation analysis of potential ecological risks of heavy metals with methane production and dewaterability in final digestate.

increased to different degrees than in the control groups. The proportions of Cu^{+}, Mn^{2+}, and Cr^{3+} increased by 11.8 %–23.1 %, 10.4 %–54.4 %, and 10.4 %–12.1 % after adding NBW, respectively (subtraction of percentages) (Fig. 7). Compared with the control, the proportion of lower energy level HMI showed an upward trend in the NBW groups. The proportions of $Ni^{2+}2p_{1/2}$, $Zn^{2+}2p_{1/2}$, and $Pb^{2+}4f_{5/2}$ increased by 4.7 %–8.7 %, 4.6 %–77.7 %, and 5.1 %–56.1 % after adding NBW, respectively (subtraction of percentages). All the above results indicate that NBW can promote the conversion of HMI in digestate to a stable valence state and energy level, making it more prone to complexation and precipitation. The local high-energy environment generated by nanobubbles during the collapse process can promote the reduction of HMI and the change of energy state, so NBW can function as a reducing agent to transform HMs into more stable and less soluble speciation [57,58].

Nanobubbles possess strong mobility and can transfer HMs to the surface of sludge particles [59]. Part of the negative charges on the surface of sludge particles were neutralized, increasing the Zeta potential, and the HMI gradually transformed from high-valence to low-valence states. Among all NBW groups, the proportions of Mn^{2+}, Cr^{3+}, $Ni^{2+}2p_{1/2}$, $Zn^{2+}2p_{1/2}$, and $Pb^{2+}4f_{5/2}$ increased the most in the A40. This is consistent with the conclusion that the A40 experienced the greatest enhancement in Zeta potential (Fig. 4b). Therefore, it can be speculated that there exists a correlation between the dewaterability and the stabilization of HMs of sludge AD. After the analysis, the HMI of all low valence states and low energy levels were negatively correlated with digestate particle size, SRF, and CST, while positively correlated with Zeta potential (Fig. 8). The HMI of all high valence states and high energy levels were positively correlated with digestate particle size, SRF, and CST, while negatively correlated with Zeta potential. Furthermore, $Zn^{2+}2p_{1/2}$ and Mn^{2+} reached the highest proportional values in the A40, 96.9 % and 89.6 %, respectively. This may be because the initial content of Zn and Mn was significantly higher than that of other HMs (Fig. 5), which was more conducive to the changes in the valence states by NBW. Correlation analysis observed that HMs with larger initial content were more significantly correlated with the changes in valence states.

Notably, methane production was positively correlated with the HMI of low valence states and low energy levels, while negatively correlated with the HMI of high valence states and high energy levels (Fig. 8). Improved methanogenesis performance was beneficial to stabilizing HMs. Further analysis revealed that the RI of HMs was significantly positively correlated with SRF and CST ($p < 0.05$), while negatively correlated with methane production (Fig. 9). Therefore, NBW may affect the stability of HMs and the potential ecological risks by changing methanogenesis and dewaterability.

3.6. Application analysis

NBW is becoming a hot spot in the field of AD, with commonly utilized gas types including CO_2, O_2, Air, N_2, and H_2 [12]. Among these options, the raw materials of Air-NBW are cost-free and have high economic value. The preparation method for Air-NBW is straightforward. Only a single instrument is needed to pump air into water, and the labor cost is low. By enhancing hydrolysis and stimulating the activity of methanogens, NBW can not only achieve a higher level of methane production and a more stable AD system but also improve the structure of flocs and significantly improve the dewaterability of digestate. Consequently, in engineering applications, the addition of NBW can reduce the dosage of flocculants needed after AD. In addition, the prepared Air-NBW is colorless and non-toxic and will not cause pollution during application, so it will not pose a threat to the environment [38]. Most importantly, Air-NBW promoted the conversion of HMs form

during AD, allowing HMs to exist within digestate in a more stable form, effectively mitigating ecological risks, and the final digestate can be used as fertilizer. To sum up, Air-NBW is a highly promising technology for engineering and land applications after sludge anaerobic treatment.

4. Conclusions

Among the three different dose groups, the 40 % NBW group (A40) exhibited the best methanogenesis and dewaterability of the final digestate, with a 16.0 % increase in methane production and a 29.8 % and 13.4 % decrease in SRF and CST, respectively, compared with the control. Additionally, NBW favored decreasing the proportion of HMs with the unstable exchangeable/acid-soluble fraction and promoted the conversion of HMI to stable low-valence states and low-energy levels in the digestate. The low-valence states and low-energy levels of HMs were positively correlated with digestate dewaterability and methane production. NBW effectively reduced the potential ecological risks of HMs, as evidenced by a 42.0 % decrease in the RI of A40 compared with the control. The RI was significantly positively correlated with SRF and CST, while negatively correlated with methane production. This study provides a novel sludge treatment strategy for controlling the pollution of HMs and efficient land use.

CRediT authorship contribution statement

Tianfeng Wang: Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition. **Ziying Xu:** Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Data curation. **Jie Wang:** Validation, Methodology, Investigation, Formal analysis. **Siqi Ren:** Investigation, Formal analysis. **Cheng Peng:** Investigation, Formal analysis. **Dan Luo:** Investigation, Formal analysis. **Xuemei Zhang:** Investigation, Formal analysis. **Miaozhi Yang:** Investigation, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jwpe.2025.108560.

Data availability

Data will be made available on request.

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